See note in red
New 8/26

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ex) $x^2+y^2=F(x,y)$

$\underline{\nabla} F = \langle 2x, 2y\rangle \neq \langle 0,0\rangle$ except $(0,0)$

Level surfaces $x^2+y^2=C$

$C = 1, 2, 3, 4$

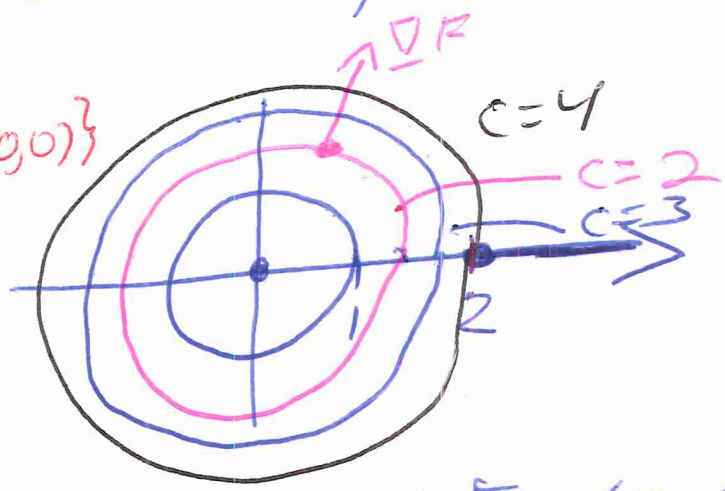
Change domain:

$\Omega = \{(x,y)\in\mathbb{R}^2 : (x,y)\neq(0,0)\}$

This is open & connected.

$\underline{\nabla} F \neq \langle 0,0\rangle$ in this domain.

Level curves do not intersect in this domain.



$(2,0)$

$\underline{\nabla} F\big|_{(2,0)} = \langle 4, 0\rangle$

What happens when $\underline{\nabla} F = (0,0)$?

not a level "curve" (point)

Property of $\underline{\nabla} F$?

① $\underline{\nabla} F$ is a vector $\perp$ to $F(x,y,z)=C$

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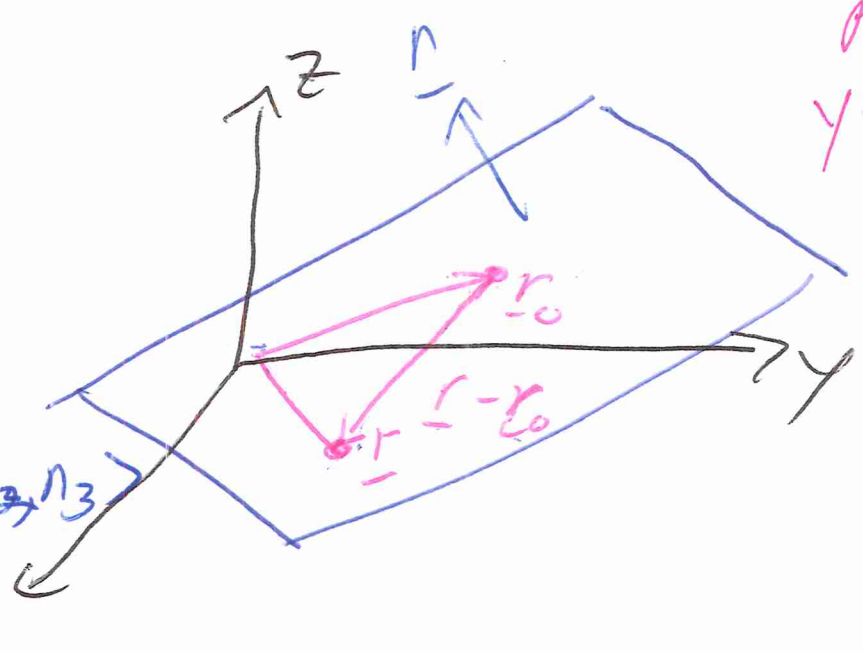
$\underline{r} = (x, y, z)$ be any other pt on plane

$\underline{r} - \underline{r}_0 =$ vector in plane

pt + m

$y - y_0 = m(x - x_0)$



$(\underline{r} - \underline{r}_0) \cdot \underline{n} = 0$ ← scaler

$\underline{n} = \langle n_1, n_2, n_3 \rangle$

$\underline{r} = (x, y, z)$

$\underline{r}_0 = (x_0, y_0, z_0)$

$(x - x_0) n_1 + (y - y_0) n_2 + (z - z_0) n_3 = 0$

$n_1 x + n_2 y + n_3 z = d$

ex) $x - y + 2z = 25$

pt. on plane $(25, 0, 0)$ $(0, -25, 0)$

$\underline{n}$ that is $\perp$ to plane

$\underline{n} = \langle 1, -1, 2 \rangle$

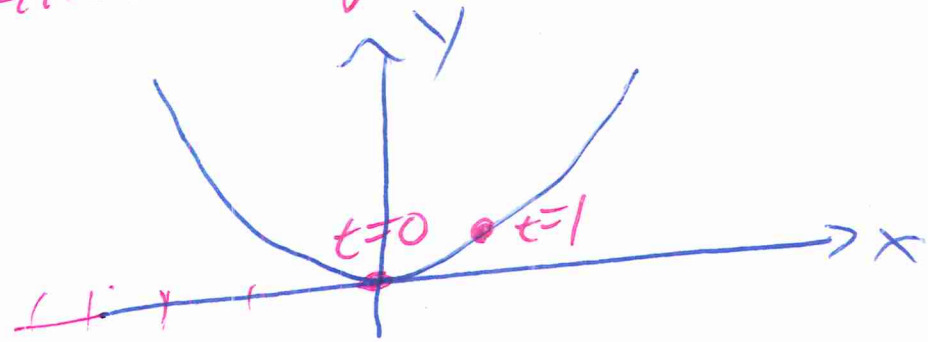
Working on changing room. Possibly spark.

51.3 Parametric representation of curves and surfaces.

• $t=-5$ $(-5,25)$

parametric: an additional variable that is called a parameter.

ex) $y=x^2$



$\underbrace{x=t,\ y=t^2}$

A parameterized eq'n. Gives more information if parameter is physical.

8-26

- fn → parameterized
  let e.g. $x = s,\ y = f(s)$
  $y = s,\ x = g(s)$

- circle or an ellipse
  $x^2 + y^2 = 9$ $\quad x = s \quad \underbrace{y = \pm\sqrt{9 - x^2}}_{\text{is not a fn}}$
  Trouble.

$\left.\begin{array}{l} x = 3\cos(t) \\ y = 3\sin(t) \end{array}\right\}$ fns :)

check: $x^2 + y^2 = 9\cos^2(t) + 9\sin^2(t) = 9$ ✓

$4x^2 + \underbrace{y^2}_{\sin^2(s)} = 1$

$= \cos^2(s) \Rightarrow x = \frac{1}{2}c$

$\langle -\frac{1}{2}\cos(s), \sin(s) \rangle$

$\langle \frac{1}{2}\cos(s), \sin(s) \rangle$

$4x^2 = \cos^2(s)$

$x^2 = \frac{1}{4}\cos^2(s)$

$x = +$ or $-$ $\quad -\frac{1}{2}\cos(s)$

parameterized → Nonparam
eliminate parameter

- $\langle t, t^2-3t+1 \rangle$

  $x=t \qquad \underline{y=x^2-3x+1}$

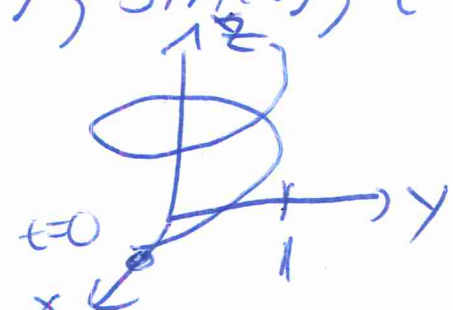
- $\langle t^2+3, t \rangle$

  $y=t \qquad \cancel{x=t^2+3} \quad x=y^2+3$

- $\langle 2\sin(t), 2\cos(t) \rangle \qquad x^2+y^2=4$

  $x=2\sin(t) \quad y=2\cos(t)$

in $\mathbb{R}^3$ $\langle \cos(t), \sin(t), t \rangle$ sketch



Parametrized surfaces

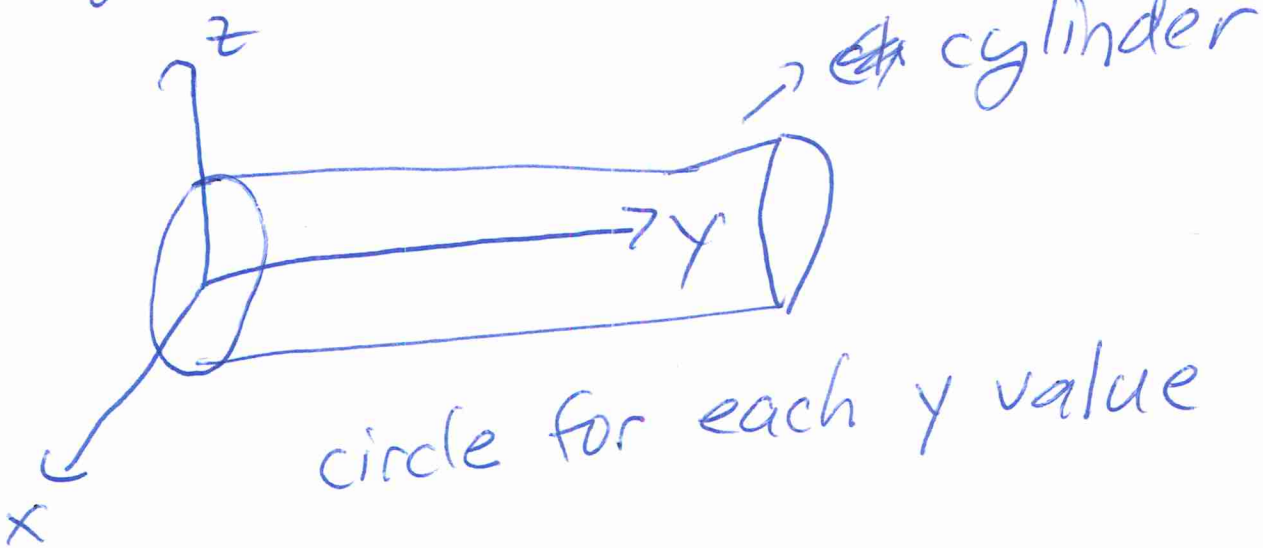
curves require one parameter, s,

surface " s 2 " s (u, v)

ex) $\langle 2\cos(u), v, 2\sin(u) \rangle$

$x^2 + z^2 = 4$ circle ~~in~~ // to xz-plane

of radius 2



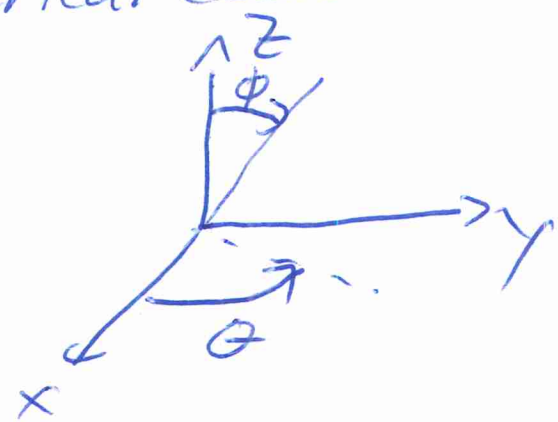
circle for each y value

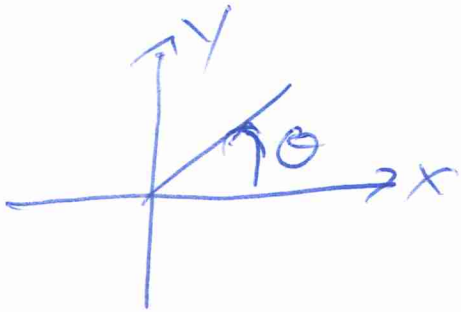
ex) Param. $x^2+y^2+z^2=a^2$

sphere of radius a

* param $= \langle a\sin\phi\cos\theta,\ a\sin\phi\sin\theta,\ a\cos\phi \rangle$

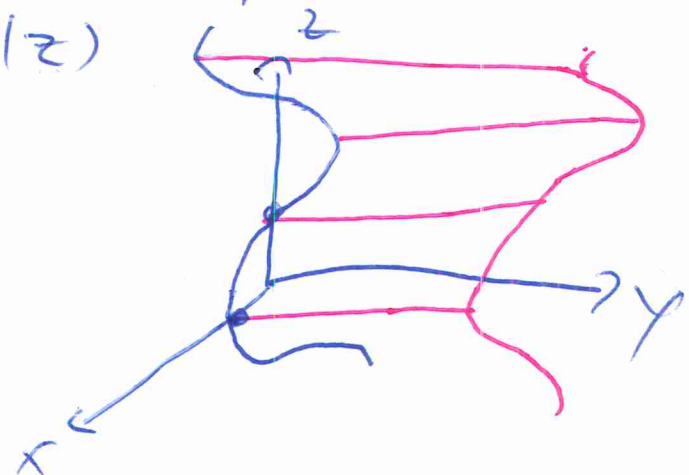
spherical coord.

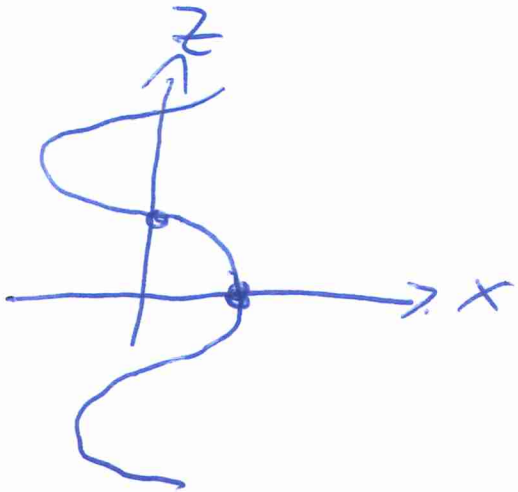




ex) $x=\cos(u),\ y=v,\ z=u$

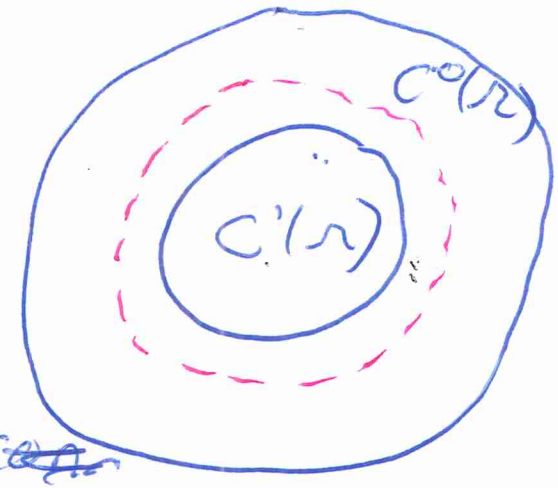
$x=\cos(z)$





51.4 ODE Misc.

Lipschitz continuous



A function $f(x)$ is Lipschitz contin. (or satisfies the Lipschitz ~~con~~ condition) if for all x_1, x_2 in a domain (an open interval) there is a constant $A > 0$ such that

$$|f(x_1) - f(x_2)| \leq A\,|x_1 - x_2|$$

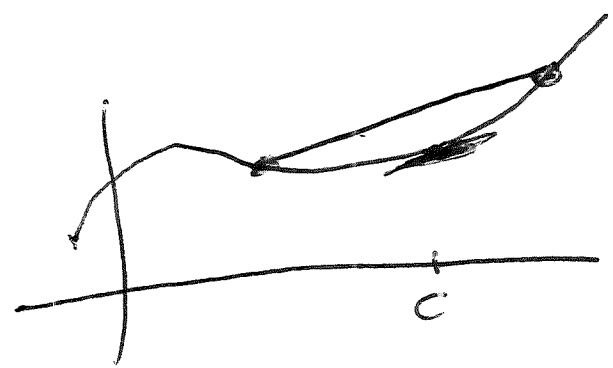
not responsible for:

<u>claim</u>: All f'ns that are $C^1(\Omega)$ are Lipschitz

By mean value Theorem holds for $C^1(\Omega)$

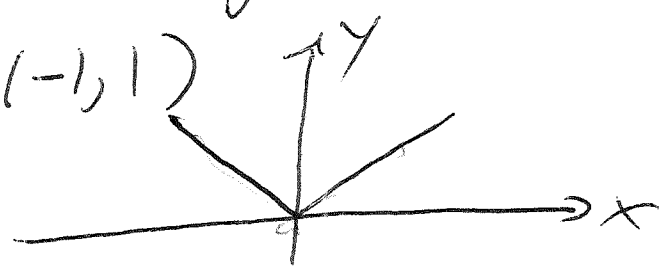
$$f(x_1) - f(x_2) = f'(c)\,(x_1 - x_2)$$

for c between x_1 + x_2



So if $f'(x) \leq A$
for all x in A
Then f is Lipschitz.

example of a f'n that is Lipschitz but not C^1: $|x|$ over $(-1,1)$



Lipschitz f'ns are C^1 f'ns with finite number of corners.

ex) Proofs in math involving PDE's are often reallge easy for boundaries that are C^1 but very difficult for boundaries with corners.